CROSS-BASE CORRELATIONS OF ARTIN PRIMES: ENTANGLEMENT, EXCLUSION, AND THE STRUCTURE OF $p-1$

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Abstract. Say a prime p is *Artin base* a if a is a primitive root modulo p. This paper studies the joint behaviour of Artin status *across bases* at the same prime. We first record a deterministic exclusion law: if $\mathrm{sqf}(ab)$ is the squarefree part of ab and c is any base with $\mathrm{sqf}(c) = \mathrm{sqf}(ab)$, then no prime is simultaneously Artin for a, b, and c — for instance, no prime has 2, 5, and 10 all as primitive roots. The proof is three lines from the multiplicativity of quadratic characters. We then measure, for all 66 pairs from the twelve bases $\{2, 3, 5, 6, 7, 10, 11, 13, 15, 17, 21, 29\}$ and all 50,847,532 primes up to 10^9, the same-prime correlation $\phi(a, b)$ between the indicator variables of Artin status. Three empirical findings emerge. First, the correlation is positive for 64 of the 66 pairs (mean $\phi = 0.0352$, individual $|z|$ up to 476); the only negative pairs, $(2, 10)$ and $(3, 15)$, are precisely pairs whose product is (a square times) another base — the entangled configurations. Second, conditioning on the number of prime factors $\omega(p-1)$ removes only 55% of the mean correlation, but conditioning on a finer signature of $p-1$ — the 2-adic valuation, the divisibility of $p-1$ by each prime up to 13, and the number of larger factors — removes 95%: the cross-base correlation is, almost entirely, a reflection of the fact that all bases see the same factorisation of $p-1$. Third, the residual that survives fine conditioning is concentrated exactly on the multiplicatively dependent pairs (mean absolute residual 0.018 within dependent pairs, versus 0.001 for multiplicatively independent pairs), and is removed completely by additionally conditioning on the quadratic-residue statuses of the two bases — identifying it as character-level entanglement, the same mechanism as the deterministic exclusion law. Finally we evaluate the joint densities predicted by the Matthews–Lenstra Kummer-degree heuristic for all 66 pairs: with no fitted parameters they match the measured densities to a mean relative error of 0.037%, and they reproduce the sign reversal on the entangled pairs. The deterministic exclusion law and the statistical residual are thereby identified as two faces of one object, the degree collapse contributed by $\mathrm{sqf}(ab)$.

1. Introduction

Artin's conjecture on primitive roots concerns one base at a time: for non-square $a \neq -1$, the set

$$\mathrm{A}_a = \{\, p \text{ prime} : a \text{ is a primitive root mod } p \,\}$$

should have density given by a rational multiple of Artin's constant, as proved under GRH by Hooley [4]. The joint distribution for *several* bases at the same prime was treated by Matthews [10], who computed (under GRH) the density of primes for which several prescribed integers are simultaneously primitive roots, and by Lenstra [8], whose Galois-theoretic framework subsumes such questions; see Moree's survey [11]. The qualitative message of that theory is that the events $p \in \mathrm{A}_a$ and $p \in \mathrm{A}_b$ are *not* independent, and that the deviation from independence is governed by the interaction of the Kummer–Galois extensions attached to a and b — in modern language, by *entanglement* between them.

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That framework remains actively developed. Järviniemi–Perucca–Sgobba [5] give a unified treatment of Artin-type problems for finitely generated groups in number fields; Kimmel [6] treats number-field and matrix variants; Klurman–Shparlinski–Teräväinen [7] obtain results on average over the base; Sgobba [12] establishes unconditional results in cases where GRH can be circumvented; and Goldmakher–Martin–Péringuey [3] propose refined conjectures on the distribution of the multiplicative order, which bear on the structure of $p-1$ that plays the central role in Section 5 below. None of this work addresses the joint statistics of Artin status across several bases at a fixed prime, which is our subject; we take [10, 8] as the source of the density predictions tested in Section 7. As those predictions are conditional on GRH, so is every comparison we make with them; our measurements themselves are of course unconditional.

This paper is the third in a series studying correlations of Artin status computationally at the scale of 10^9 [1, 2]. The first paper [1] measured the correlation between the Artin statuses (base 10) of *consecutive* primes and decomposed it into residue channels; the second proved a classification of deterministic exclusion laws for consecutive Artin primes in arbitrary bases and identified the conductor of the quadratic character χ_a as the parameter governing the statistical correlation. Here we complete the picture along the remaining axis: *fix the prime, vary the base.*

Two questions organise the paper.

(Q1) What is deterministically forbidden? For consecutive primes, the answer [2] was a set of gap classes on which the quadratic character is forced to flip. For a single prime and several bases, the answer is an exclusion law inside *multiplicative triples*:

Theorem 1 (Same-prime exclusion law; see Theorem 2)**.** *Let a, b be non-square positive integers and let c be any non-square positive integer with* $\operatorname{sqf}(c) = \operatorname{sqf}(ab)$*. Then no prime $p \nmid 2abc$ is simultaneously Artin base a, Artin base b, and Artin base c.*

For example, no prime has all of $2, 5, 10$ as primitive roots; no prime has all of $3, 5, 15$; and, since sqf is all that matters, no prime has all of $2, 5, 40$. Like the gap exclusion law of [2], the statement is elementary — it needs only the multiplicativity of the quadratic character and the fact that a quadratic residue is not a primitive root — and, like that law, we have not found it stated explicitly in the literature, although it is certainly implicit in the density computations of [10, 8], where the entangled character relation $\chi_a\chi_b = \chi_{ab}$ is exactly what produces degenerate factors.

(Q2) What does the joint distribution actually look like, and what explains it? Here the data speak. We compute the exact 2×2 contingency table of $\big(\mathbf{1}[p \in \mathrm{A}_a],\, \mathbf{1}[p \in \mathrm{A}_b]\big)$ over all 50,847,532 primes $p \le 10^9$, for all 66 pairs from the base set

$$\{2, 3, 5, 6, 7, 10, 11, 13, 15, 17, 21, 29\},$$

chosen to contain many multiplicative triples (15 of the 66 pairs lie in a triple $\operatorname{sqf}(ab) = \operatorname{sqf}(c)$ within the set) alongside multiplicatively independent pairs. The findings, developed in Sections 4–6:

(1) *Positivity.* $\phi(a, b) > 0$ for 64 of 66 pairs, with mean 0.0352 and z-scores up to 476: knowing that a is a primitive root mod p makes it *more* likely that b is. The largest correlations all involve base 5 (e.g. $\phi(5, 13) = 0.0667$: $P(\mathrm{A}_{13} \mid \mathrm{A}_5) = 0.416$ against $P(\mathrm{A}_{13} \mid \neg\mathrm{A}_5) = 0.350$).

(2) *The exceptions are the entangled pairs.* The only two negative correlations are $\phi(2, 10) = -0.0302$ and $\phi(3, 15) = -0.0303$ — precisely the pairs (a, b) in our set with $b = a \cdot c$ for another base c of the set (up to squares). Multiplicative dependence reverses the sign of the association.

(3) *The positive correlation is $p-1$ structure, not entanglement.* Conditioning on $\omega(p-1)$ removes 55% of the mean correlation; conditioning on a fine signature of $p-1$ — namely $\big(\min(v_2(p-1),4),$ the set of primes $q \le 13$ dividing $p-1$, $\min(\omega_{>13}(p-1),3)\big)$ — removes 95%, leaving a mean residual of 0.0019. All bases are tested against the same integer $p-1$; a prime whose $p-1$ is "primitive-root friendly" (few small factors, small 2-part) is friendlier for *every* base simultaneously. This common-cause structure, not any interaction between the bases, is the bulk of the correlation.

(4) *What survives is entanglement, and it is exactly localised.* After fine conditioning, the residual correlation among multiplicatively *dependent* pairs (mean $|\phi_{\mathrm{sig}}| \approx 0.005$, and mean absolute residual 0.0179, with individual values reaching -0.034 for $(2,10)$ and -0.026 for $(5,15)$) is eighteen times that of independent pairs (0.0010); and additionally conditioning on the quadratic-residue statuses $(\chi_a(p), \chi_b(p))$ removes it entirely, flattening dependent and independent pairs to a common $+0.005$. The residual is thus identified, empirically, as the quadratic-character entanglement $\chi_a\chi_b = \chi_{\mathrm{sqf}(ab)}$ — the statistical shadow of Theorem 2.

(5) *Theory reproduces all of it.* Evaluating the Matthews–Lenstra Kummer-degree heuristic (2) for each pair (Section 7) predicts the measured joint densities to a mean relative error of 0.037% over the 66 pairs, with no fitted parameters, including the negative ϕ of the two entangled pairs. The entangled degree collapse in the model is precisely the character relation behind Theorem 2.

The theme of the series persists: at every level, the joint behaviour of Artin primes decomposes into a small deterministic core (here, the triple exclusion law) plus a statistical component with an identifiable arithmetic carrier (here, the shared factorisation of $p-1$, with entanglement as a measurable second-order correction).

2. The same-prime exclusion law

Throughout, for a non-square positive integer a we write $d_a = \mathrm{sqf}(a)$ for its squarefree part and $\chi_a = \big(\frac{d_a}{\cdot}\big)$ for the quadratic character attached to $\mathbb{Q}(\sqrt{a})$; thus $\chi_a(p) = +1$ exactly when a is a quadratic residue modulo p (for $p \nmid 2a$), and

$$p \in \mathrm{A}_a \implies \chi_a(p) = -1, \tag{1}$$

since a quadratic residue generates a subgroup of index at least 2 in $(\mathbb{Z}/p\mathbb{Z})^\times$.

Theorem 2 (Triple exclusion)**.** *Let a, b, c be non-square positive integers with $\mathrm{sqf}(c) = \mathrm{sqf}(ab)$, and let $p \nmid 2abc$ be prime. Then p is not simultaneously Artin base a, Artin base b, and Artin base c.*

Proof. For any prime $p \nmid 2ab$ one has, by complete multiplicativity of the Legendre symbol in the numerator, $\chi_a(p)\,\chi_b(p) = \big(\frac{d_a d_b}{p}\big) = \big(\frac{\mathrm{sqf}(ab)}{p}\big) = \chi_c(p)$, since $d_a d_b$ and $\mathrm{sqf}(ab)$ differ by a square. If p were Artin for all three bases, then by (1) all three character values would be -1, giving $(-1)(-1) = -1$, a contradiction. □

Corollary 3. *Within any multiplicative triple — for instance* $(2,5,10)$*,* $(3,5,15)$*,* $(3,7,21)$*,* $(2,3,6)$ *— a prime can be Artin for at most two of the three bases.*

Remark 4. The exclusion is sharp: primes Artin for exactly two of $(2,5,10)$ exist in abundance. The smallest is $p = 7$, where 5 and 10 are both primitive roots (each of order 6) while 2 has order 3; likewise $p = 23$, where 5 and 10 are primitive roots and 2 is a quadratic residue of order 11. More systematically, in our data the pair $(2,10)$ has $P(\mathrm{A}_{10} \mid \mathrm{A}_2) = 0.355$: two-of-three is common, three-of-three is impossible.

Remark 5. Theorem 2 is the same-prime analogue of the gap exclusion law for consecutive primes proved in [2]: both are instances of a parity obstruction in the group generated by the quadratic characters involved. There the two characters were χ_a evaluated at p and at $p+g$, and the obstruction was arranged by the shift g; here the characters belong to different bases at the same prime, and the obstruction is arranged by multiplicative dependence. The Galois-theoretic density computations of Matthews [10] and Lenstra [8] encode this relation implicitly (the degree of the compositum of the relevant Kummer extensions drops); the pointwise statement seems worth recording.

3. Data and computation

By a segmented sieve we enumerated all primes $p \le 10^9$ (50,847,534 primes, of which 50,847,532 satisfy $p \nmid 2 \cdot (\text{base})$ for all twelve bases and enter the analysis). For each prime we factored $p-1$ once by trial division and computed the Artin indicator for each of the twelve bases by the standard criterion ($a^{(p-1)/\ell} \not\equiv 1 \bmod p$ for every prime $\ell \mid p-1$, in 128-bit arithmetic). One streaming pass accumulates, for each of the 66 unordered base pairs, the 2×2 contingency table of joint Artin status — globally, and stratified by (i) $\omega(p-1)$ and (ii) the fine signature

$$\mathrm{sig}(p) = \Big(\min(v_2(p-1), 4),\ \ \{\, q \le 13 : q \mid p-1 \,\},\ \ \min(\omega_{>13}(p-1), 3)\Big),$$

a partition of the primes into at most $5 \cdot 32 \cdot 4 = 640$ cells. The signature records exactly the information about $p-1$ that the Artin criterion consults for small bases: the 2-part and the small odd prime divisors. A separate run, also to 10^9 (50,847,524 primes), stratified each pair additionally by the quadratic-residue statuses $\big(\chi_a(p), \chi_b(p)\big)$ of the two bases on top of the signature. The whole computation runs in under two hours on a single core of a commodity machine (2-core Intel Xeon E5-2673 v4, 8 GB RAM); all counts are exact.

As validation, the marginal Artin densities matched the Hooley densities to four decimal places for every base — e.g. 0.373993 for the bases with squarefree part $\not\equiv 1 \pmod 4$, against Artin's constant 0.373956, and 0.393642 for base 5, whose density carries the classical correction factor $C \cdot \frac{20}{19}$ for $d \equiv 1 \pmod 4$ (see [4, 11]).

4. The cross-base correlation

For each pair write $\phi(a,b)$ for the Pearson correlation of the indicator variables (the ϕ coefficient of the 2×2 table). At $N \approx 5.08 \times 10^7$ primes, the standard error of ϕ under independence is $N^{-1/2} \approx 0.00014$; as in the previous papers of the series, the enormous resulting z-scores certify only that sampling noise is negligible, and the relevant quantity is the effect size.

Three observations, in increasing order of depth.

Observation 6 (Positivity)**.** $\phi(a,b) > 0$ for 64 of the 66 pairs. Artin statuses of different bases at the same prime *attract.*

The sign is the opposite of the consecutive-prime correlation measured in the first two papers of the series [1, 2] (where $\delta < 0$ uniformly), and the mechanism is transparent: there the two primes had *different* (and LOS-anticorrelated) values of $p-1$; here every base is tested against the *same* $p-1$. If $p-1$ has few small prime factors and a small 2-part, the primitive-root condition is easier for every base at once. Formally, the Artin indicators share the common cause $\mathrm{sig}(p)$.

TABLE 1. Same-prime cross-base Artin correlation at $x = 10^9$: summary and extremes. "Dependent" means the pair lies in a multiplicative triple within the base set ($\mathrm{sqf}(ab) = \mathrm{sqf}(c)$).

	pairs	mean ϕ
all pairs	66	$+0.0352$
multiplicatively independent	51	$+0.0363$
multiplicatively dependent	15	$+0.0314$
largest: $(5, 13)$		$+0.0667$
smallest: $(3, 15)$		-0.0303
second smallest: $(2, 10)$		-0.0302

TABLE 2. The ten largest $|\phi|$ at $x = 10^9$, with the values after conditioning on $\omega(p-1)$ and on the fine signature $\mathrm{sig}(p)$ (pair-weighted mean of within-cell ϕ; cells with fewer than 200 primes excluded). "Dep." marks multiplicatively dependent pairs.

a	b	ϕ	$\phi \mid \omega$	$\phi \mid \mathrm{sig}$	Dep.
5	13	$+0.0667$	$+0.0200$	$+0.0003$	
5	17	$+0.0649$	$+0.0218$	$+0.0025$	
5	29	$+0.0631$	$+0.0246$	$+0.0007$	
5	10	$+0.0624$	$+0.0285$	-0.0125	✓
5	6	$+0.0623$	$+0.0292$	$+0.0005$	
5	11	$+0.0623$	$+0.0300$	$+0.0004$	
5	15	$+0.0622$	$+0.0297$	-0.0255	✓
3	5	$+0.0622$	$+0.0284$	$+0.0015$	✓
5	7	$+0.0622$	$+0.0297$	$+0.0002$	
2	5	$+0.0621$	$+0.0336$	$+0.0004$	✓

Observation 7 (The negative pairs are the entangled ones)**.** The only pairs with $\phi < 0$ are $(2, 10)$ and $(3, 15)$: in each, the product of the two bases is, up to a square, the third base of a multiplicative triple contained in the set ($2 \cdot 10 = 4 \cdot 5$, $3 \cdot 15 = 9 \cdot 5$). For $(2, 10)$: $P(\mathrm{A}_{10} \mid \mathrm{A}_2) = 0.3551$ versus $P(\mathrm{A}_{10} \mid \neg\mathrm{A}_2) = 0.3853$.

The sign reversal has a character-theoretic origin. For a dependent pair, $\chi_a\chi_b = \chi_5$ (in both our examples), so on the half of primes with $\chi_5(p) = +1$ the values $\chi_a(p)$ and $\chi_b(p)$ are forced *equal* — fine for joint Artin status — but on the half with $\chi_5(p) = -1$ they are forced *opposite*, and then at most one of the two bases can be Artin: a pointwise exclusion on half the primes, diluted into the observed negative correlation. This is Theorem 2 operating statistically through the third character.

5. Decomposition: the correlation is the structure of $p - 1$

Conditioning on $\omega(p-1)$ alone — the crude size of the factorisation — removes barely half of the correlation. This is worth emphasising because ω is the statistic through which the consecutive-prime papers routed the mechanism, and it is genuinely insufficient here: two primes with the same $\omega(p-1)$ can present very different Artin conditions (it matters *which*

Table 3. Mean cross-base correlation after conditioning, all 66 pairs at $x = 10^9$.

conditioning	mean ϕ	removed
none	0.0352	—
$\omega(p-1)$	0.0158	55%
sig(p): $\big(v_2,\ \{q \le 13\},\ \omega_{>13}\big)$	0.0019	95%

Table 4. Residual correlation after conditioning on sig(p) and after additionally conditioning on the pair of quadratic-residue statuses $(\chi_a(p), \chi_b(p))$, split by multiplicative dependence. All values at $x = 10^9$ over 50,847,524 primes.

group	mean ϕ	mean $\phi \mid$ sig	mean $\phi \mid$ sig, QR
multiplicatively dependent (15 pairs)	+0.0314	+0.0049	+0.0049
mean *absolute* residual		0.0179	
multiplicatively independent (51 pairs)	+0.0363	+0.0010	+0.0052
mean *absolute* residual		0.0010	
all 66 pairs	+0.0352	+0.0019	+0.0052

primes divide $p-1$, and with what 2-adic valuation, not how many). The fine signature — 2-adic valuation capped at 4, the exact set of primes ≤ 13 dividing $p-1$, and the count of larger prime factors capped at 3 — is the information the Artin criterion actually consults for small bases, and conditioning on it removes 95% of the mean correlation (Table 3). That the fine structure of $p-1$, rather than its coarse size, is what governs Artin behaviour is consistent with the refined order-distribution conjectures of [3], where the same distinction between $\omega(p-1)$ and the actual divisor structure is what refines Artin's original heuristic.

We regard this as the main statistical finding: *cross-base Artin correlation at a single prime is, to first order, not an interaction between the bases at all.* It is a common-cause effect of the shared integer $p-1$. Any model that treats A_a and A_b as conditionally independent given the small-prime structure of $p-1$ reproduces 95% of the observed association.

6. The residual is entanglement

The 5% that survives is not noise, and it is not uniformly distributed: it sits precisely on the multiplicatively dependent pairs.

Two facts pin down the mechanism, both measured over the full 10^9 range.

First, after fine conditioning the dependent pairs retain substantial residuals of the sign predicted by the character relation of Observation 7: their mean *absolute* residual is 0.0179, eighteen times the 0.0010 of the 51 independent pairs. Indeed the six most negative residuals in the entire collection are *exactly* six dependent pairs and nothing else:

$$(2,10)\colon -0.0343, \quad (5,15)\colon -0.0255, \quad (6,10)\colon -0.0144, \quad (5,10)\colon -0.0125, \quad (7,21)\colon -0.0060, \quad (10,15)\colon -0.0$$

Note that $(2,10)$ and $(5,15)$ are more negative *after* conditioning than before, and that $(5,10)$, $(5,15)$, $(6,10)$, $(10,15)$ and $(7,21)$ change sign under conditioning: the common-cause channel had been masking an underlying repulsion.

Second, additionally conditioning on the pair of quadratic-residue statuses $(\chi_a(p), \chi_b(p))$ removes this residual completely. Every one of the six negative values above becomes positive and small (+0.0031, +0.0066, +0.0045, +0.0065, +0.0040, +0.0048 respectively), and the dependent and independent groups become indistinguishable at $+0.0049$ and $+0.0052$ (Table 4). That common small positive floor of about $+0.005$ is present for all 66 pairs and is plausibly attributable to higher-order (cubic and quartic) conditions shared through $p-1$ beyond what our signature records.

Conditioning on the quadratic characters is precisely what removes character-level entanglement; that it flattens exactly the pairs with $\chi_a\chi_b = \chi_{\mathrm{sqf}(ab)}$, and only those, identifies the residual as the statistical trace of the entangled relation — the same object that makes Theorem 2 true.

The picture that emerges is cleanly layered:

$$\underbrace{\text{common cause } p-1}_{\approx 95\%} + \underbrace{\text{quadratic entanglement } \chi_a\chi_b = \chi_{\mathrm{sqf}(ab)}}_{\text{residual, sign-definite, localised}} + \underbrace{\text{higher-order sharing}}_{\text{small positive floor}}.$$

7. Comparison with the predicted joint densities

The measurements of Sections 4–6 identify the mechanisms empirically. We now check them against theory quantitatively, by computing the joint densities predicted by the Matthews–Lenstra framework and comparing with the observed counts pair by pair.

Write $d_a = \mathrm{sqf}(a)$. The heuristic, in the form given by [10] and systematised in [8], is an inclusion –exclusion over the degrees of the Kummer extensions that detect the conditions "a is an ℓ-th power residue":

$$\mathbf{P}\big(p \in \mathrm{A}_a \cap \mathrm{A}_b\big) \;=\; \sum_{m,n \text{ squarefree}} \frac{\mu(m)\,\mu(n)}{\big[\mathbb{Q}(\zeta_L, a^{1/m}, b^{1/n}) : \mathbb{Q}\big]}, \qquad L = \mathrm{lcm}(m,n), \tag{2}$$

and the entanglement enters through the degree in the denominator. Generically $[\mathbb{Q}(\zeta_L, a^{1/m}, b^{1/n}) : \mathbb{Q}] = \varphi(L)\,m\,n$, but the degree *drops by a factor of* 2 *for each quadratic class in the group*

$$V(m,n) = \big\langle\, d_a \ (\text{if } 2 \mid m),\ d_b \ (\text{if } 2 \mid n) \,\big\rangle \subseteq \mathbb{Q}^\times/(\mathbb{Q}^\times)^2$$

whose quadratic field lies inside $\mathbb{Q}(\zeta_L)$, i.e. whose conductor divides L. The crucial point for this paper is that when both $2 \mid m$ and $2 \mid n$ the group V contains not only d_a and d_b but their product $d_a d_b \sim \mathrm{sqf}(ab)$: the entangled third character of Theorem 2 appears in the model as an extra degree collapse, exactly for the multiplicatively dependent pairs. We evaluate (2) by summing the ramified part exactly over all squarefree m, n supported on $S = \{2\} \cup \{q : q \mid d_a d_b\}$ and folding the rest into the Euler product $\prod_{q \notin S}\big(1 - \frac{2q-1}{q^2(q-1)}\big)$, truncated at 3×10^6.

As a check on the implementation, the same code specialised to a single base reproduces the classical Hooley densities: 0.373956 for bases with $d \not\equiv 1 \pmod 4$ (Artin's constant), 0.393638 for $d = 5$, 0.376368 for $d = 13$, against measured 0.373993, 0.393642, 0.376369.

Observation 8 (The model reproduces the measurements)**.** Over all 66 pairs at $x = 10^9$, the mean absolute relative error between the predicted and measured joint densities is 0.037%, with maximum 0.67% (attained at $(5, 21)$). In particular the model predicts the sign reversal: for the entangled pairs $(2, 10)$ and $(3, 15)$ it gives $j = 0.132776$ and $\phi = -0.0302$, against measured $j = 0.132776$, 0.132769 and $\phi = -0.0303$ for both.

This is the quantitative counterpart of Sections 4 and 6, and it settles the interpretation. The 66-pair pattern — positive correlation almost everywhere, negative precisely on the two entangled pairs, with magnitudes ordered as observed — is not merely *consistent with* the Matthews–Lenstra framework; it is reproduced by it to within a few parts in 10^4, with no fitted parameters. Correspondingly, the deterministic law of Theorem 2 and the statistical residual

Table 5. Predicted versus measured joint Artin densities, selected pairs at $x = 10^9$. "Dep." marks multiplicatively dependent pairs. The final two rows are the two negatively correlated pairs.

a	b	measured j	model j	rel. err.	measured ϕ	model ϕ	Dep.
5	13	0.163939	0.164671	+0.45%	+0.0667	+0.0698	
5	10	0.161962	0.161922	−0.02%	+0.0624	+0.0623	✓
5	15	0.161934	0.161922	−0.01%	+0.0622	+0.0623	✓
3	7	0.149965	0.149971	+0.00%	+0.0432	+0.0433	✓
13	17	0.150249	0.150314	+0.04%	+0.0383	+0.0386	
2	3	0.147361	0.147349	−0.01%	+0.0321	+0.0321	✓
2	6	0.147334	0.147349	+0.01%	+0.0320	+0.0321	✓
7	21	0.144754	0.144728	−0.02%	+0.0238	+0.0238	✓
15	21	0.144725	0.144728	+0.00%	+0.0237	+0.0238	
3	15	0.132769	0.132776	+0.01%	−0.0303	−0.0302	✓
2	10	0.132776	0.132776	+0.00%	−0.0303	−0.0302	✓

isolated in Section 6 are two faces of one object: the degree collapse contributed by $\mathrm{sqf}(ab)$ in $V(m, n)$.

Remark 9. The largest discrepancies (0.67% at $(5, 21)$, 0.11% at $(13, 21)$) involve base 21, the only base in our set whose squarefree part has two odd prime factors; the residual is consistent with the truncation of the Euler tail at 3×10^6 interacting with the larger ramified set $S = \{2, 3, 5, 7\}$, and we have not pursued it. Ordinary Hooley-type errors of this size are also expected at $x = 10^9$, since the convergence in Hooley's theorem is only logarithmic.

8. Discussion

Relation to the first two papers of the series [1, 2]. The trilogy now covers both axes of the joint distribution of Artin status. Along the prime axis (consecutive p_n, p_{n+1}, one base), correlation is negative, driven by the Lemke Oliver–Soundararajan bias [9] propagating through the two distinct integers $p_n - 1$ and $p_{n+1} - 1$, with deterministic exclusion classes in the gap. Along the base axis (one prime, two bases), correlation is positive, driven by the single shared integer $p - 1$, with deterministic exclusion inside multiplicative triples. In both cases the deterministic law is a parity statement about quadratic characters, and in both cases it is statistically dominated by a factorisation-of-$p-1$ mechanism — diluted to invisibility in the global average there, absorbed almost entirely by conditioning here.

Future work. (1) A rigorous version of Section 7: the comparison there is between measurement and a heuristic evaluated numerically, and the error terms — both the Hooley-type error at $x = 10^9$ and the truncation of our Euler tail — are not controlled. Making the agreement into a theorem (conditionally on GRH, in the manner of [10]) would require uniform error bounds we do not attempt. (2) The mixed axis: ϕ between $\mathrm{A}_a(p_n)$ and $\mathrm{A}_b(p_{n+1})$ for $a \neq b$ — the data pipeline extends immediately, and the prediction from the present findings is a small negative correlation governed by the two conductors jointly. (3) Higher-order entanglement: triples of bases conditioned on quadratic data should expose the cubic layer ($\ell = 3$ Kummer conditions), with $\mathbb{Q}(\zeta_3)$ playing the role $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{2})$ played here.

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Data availability

All code and results are publicly available at https://github.com/jpbald93/consecutive-artin, including the single-pass multi-base programs (`crossbase.c`, `crossbase_fine.c`, `crossbase_qr.c`), the exact contingency tables underlying every value quoted here, and the analysis scripts. The full computation reproduces in under two hours on a single core.

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